

Paired Prompt-Contrastive Training Restores Prompt Faithfulness in 3D Medical Segment Anything Models

Anonymous AAAI Submission

Abstract

3D medical Segment Anything Models promise interactive segmentation via point prompts, yet we discover they suffer from *prompt unfaithfulness*: the model largely ignores prompt location and predicts based on image priors alone, achieving only 20.8% Switch Accuracy—the rate at which predictions correctly change when prompts target different organs. This failure is distinct from prompt robustness (only 1.3-point Dice drop under 10-voxel noise) and is hidden by standard Dice evaluation. We identify the root cause as single-prompt training, which never requires the model to produce different outputs for different prompts on the same image. To address this, we propose *Paired Prompt-Contrastive Training*: for each image, we sample two organ prompts and supervise both predictions through a shared LoRA-adapted decoder. Comprehensive mechanism ablation reveals that paired prompt exposure itself is the dominant factor (+40 points Switch Accuracy), while auxiliary losses (grounding regularization, switch margin) provide only marginal additional benefit (+0–5 points). Our method achieves Switch Accuracy 0.993 ± 0.012 (from 0.208 baseline) with only 4.4-point Dice cost on AMOS across three seeds, generalizes to BTCV external validation (Switch 1.000), and adds merely 5.3% training overhead with zero inference cost. Code will be released upon acceptance.

1 Introduction

Segment Anything Models (SAM) adapted for 3D medical imaging [Wang et al., 2023, Du et al., 2023, Ma et al., 2024, Zhu et al., 2024] enable interactive organ segmentation through point prompts: a clinician clicks on a target organ, and the model produces a volumetric segmentation mask. This paradigm assumes a fundamental contract—changing the prompt target should change the predicted mask. The promise of such models lies precisely in this interactive controllability, which enables efficient annotation workflows and intraoperative guidance.

We discover that this contract is systematically violated. When evaluated on multi-organ abdominal CT, SAM-Med3D [Wang et al., 2023] achieves only 20.8% *Switch Accuracy*—the rate at which the model correctly changes its prediction when the prompt moves from one organ to another (Figure 1). The model predicts the same dominant organ regardless of prompt location, effectively ignoring user input. This means that in nearly 80% of cases, clicking on a different organ produces the same segmentation output—a catastrophic failure for interactive clinical use.

This failure is *not* a robustness problem. Recent work [Chatopadhyay et al., 2026] studies sensitivity to prompt perturbations, finding that models degrade under noise. In contrast,

SAM-Med3D is remarkably robust to prompt noise (only 1.3-point Dice drop at 10-voxel perturbation). The failure we identify is a *controllability* problem: the model has learned strong anatomical priors that override prompt information entirely. Standard Dice evaluation completely hides this problem because Dice rewards accurate segmentation regardless of whether the model actually follows the prompt.

We trace the root cause to single-prompt training. When each training step presents only one prompt per image, the model can achieve low segmentation loss by exploiting image priors—always predicting the largest or most salient organ—without learning prompt-conditioned behavior. The model never needs to demonstrate that different prompts produce different outputs. This is analogous to a language model that produces the same response regardless of the input query: technically functional but fundamentally uncontrollable.

Our fix is conceptually simple: *Paired Prompt-Contrastive Training*. For each training image, we sample two organ prompts, compute predictions for both using a shared decoder, and supervise both outputs. This forces the model to produce different masks for different prompts on the same image—it can no longer satisfy the training objective by ignoring prompts. Through comprehensive mechanism ablation, we discover that this paired exposure is the dominant factor—achieving +40 points Switch Accuracy improvement—while auxiliary losses (grounding regularization, switch margin) provide only marginal additional benefit (+0–5 points).

Our contributions are:

1. We identify *prompt unfaithfulness* as a distinct failure mode in 3D medical SAM models, measurable via Switch Accuracy, and show it is hidden by standard Dice evaluation yet catastrophic for interactive use.
2. We propose Paired Prompt-Contrastive Training and demonstrate through factor-isolation ablation that paired prompt exposure—not any specific loss function—is the core mechanism restoring faithfulness.
3. We validate across datasets (AMOS, BTCV) with 3-seed experiments, achieving Switch Accuracy 0.993 ± 0.012 with Dice 0.676 ± 0.006 , at only 5.3% training overhead and zero inference cost.
4. We show that prompt unfaithfulness is specific to 3D medical SAM adaptation (SAM2 achieves 0.954 Switch on natural images), suggesting it arises from limited medical training data and strong anatomical priors.

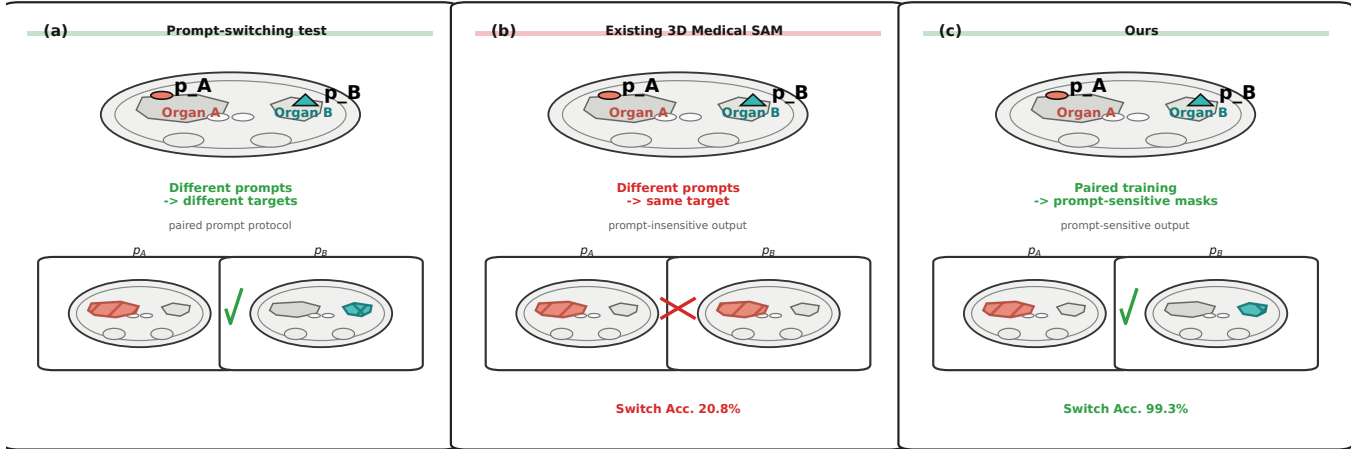


Figure 1: Prompt unfaithfulness in 3D medical SAM. (a) Expected behavior: different point prompts should produce different segmentation masks. (b) Observed behavior: SAM-Med3D produces nearly identical predictions regardless of prompt target (Switch Accuracy = 20.8%). (c) After our Paired Prompt-Contrastive Training: the model correctly switches predictions to match the prompted organ (Switch Accuracy = 99.3%).

2 Related Work

Foundation Models for 3D Medical Segmentation. The Segment Anything Model (SAM) [Kirillov et al., 2023] introduced promptable segmentation for natural images. SAM2 [Ravi et al., 2024] extended this to video with memory-augmented tracking. Several works adapt these models to 3D medical imaging: SAM-Med3D [Wang et al., 2023] trains a volumetric SAM on large-scale CT data with point prompts; MedSAM [Ma et al., 2024] fine-tunes SAM for medical bounding-box prompts; Medical SAM2 [Zhu et al., 2024] reframes 3D segmentation as video tracking via SAM2; and MA-SAM [Chen et al., 2024a] proposes modality-agnostic adapters for cross-modal transfer. SAM2-Adapter [Chen et al., 2024b] introduces task-specific adapters to overcome SAM2 limitations in medical and camouflage domains. These works evaluate segmentation accuracy (Dice) but do not assess whether the model faithfully follows prompt targets—a gap our work addresses.

Parameter-Efficient Adaptation. LoRA [Hu et al., 2022] enables efficient fine-tuning by injecting low-rank matrices into frozen models. Recent medical imaging works apply parameter-efficient methods extensively: LoRA-MedSAM [Wu, 2024] reduces fine-tuning cost while improving Dice; FATE-SAM [Deng, 2025] achieves few-shot 3D segmentation by reassembling SAM2 modules without training; and tCUR-LoRA [Cheng et al., 2025] proposes tensor decomposition for more efficient adaptation. Our work uses standard LoRA but reveals that the *training paradigm* (paired vs. single prompt) matters far more than the adaptation architecture.

Prompt Robustness in Medical Segmentation. Chattopadhyay et al. [2026] systematically evaluate how 3D medical segmentation models respond to imprecise visual prompts, revealing reliance on spatial cues. Huang et al. [2024] study

perturbed prompts for robust SAM adaptation in polyp segmentation. Li [2024] propose saliency-guided prompt distillation for robust segmentation under noisy prompts. These works study *robustness*—stability under small perturbations. We identify a fundamentally different failure: *faithfulness*—whether the model responds to target changes at all. SAM-Med3D is robust (1.3pt Dice drop at shift-10) but unfaithful (Switch Accuracy 20.8%)—these are orthogonal properties requiring different solutions.

Contrastive and Paired Training in Segmentation. Contrastive learning has been widely applied to medical image analysis [Zhao, 2024]. Recent works address dimensional collapse in medical contrastive pre-training [Liu, 2024] and propose domain-focused example selection for data-efficient segmentation [Wang, 2025]. Consistency regularization enforces prediction stability across augmented views. Our paired prompt training shares the principle of exposing the model to contrasting conditions, but uniquely targets prompt-conditioned output *differentiation*—forcing different outputs for different prompts—rather than representation alignment or augmentation invariance.

3 Method

3.1 Problem Formulation

Let f_θ be a promptable segmentation model that takes a 3D image $\mathbf{x} \in \mathbb{R}^{D \times H \times W}$ and a point prompt $p = (d, h, w)$ to produce a binary mask $f_\theta(\mathbf{x}, p) \in \{0, 1\}^{D \times H \times W}$. We define *prompt faithfulness* as the property that changing the prompt target changes the prediction accordingly.

Switch Accuracy. Given an image $\mathbf{x}$ containing organs A and B with ground-truth masks y_A and y_B , and prompts $p_A \in A, p_B \in B$, we define a *switch* as successful if both

conditions hold simultaneously:

$$\text{Dice}(f_\theta(\mathbf{x}, p_A), y_A) > \text{Dice}(f_\theta(\mathbf{x}, p_A), y_B) \quad (1)$$

$$\text{Dice}(f_\theta(\mathbf{x}, p_B), y_B) > \text{Dice}(f_\theta(\mathbf{x}, p_B), y_A) \quad (2)$$

Switch Accuracy is the fraction of paired prompts satisfying both conditions across a fixed evaluation bank. This metric captures whether the model produces the *correct* organ for each prompt, not merely whether it produces *something*.

3.2 Why Standard Training Fails

Standard training optimizes $\mathcal{L}_{\text{seg}}(f_\theta(\mathbf{x}, p), y)$ independently for each prompt. This objective is satisfiable without prompt faithfulness: a model that always predicts the largest organ achieves reasonable Dice regardless of prompt location. Formally, if organ A dominates the volume, then $f_\theta(\mathbf{x}, p) \approx y_A$ for all p achieves high average Dice without ever using p . The loss contains no term enforcing that $f_\theta(\mathbf{x}, p_A) \neq f_\theta(\mathbf{x}, p_B)$ when $A \neq B$.

This explains why standard LoRA fine-tuning (seg-only) improves Dice from 0.450 to 0.720 but Switch Accuracy only reaches 0.483—the model learns better segmentation but not better prompt following.

3.3 Paired Prompt-Contrastive Training

We address this by training with paired prompts (Figure 2). For each image $\mathbf{x}$ containing organs A and B :

1. Compute image embedding $\mathbf{e} = \text{Enc}(\mathbf{x})$ once (frozen encoder)
2. Decode with prompt p_A : $\hat{y}_A = \text{Dec}_\phi(\mathbf{e}, p_A)$
3. Decode with prompt p_B : $\hat{y}_B = \text{Dec}_\phi(\mathbf{e}, p_B)$
4. Supervise both predictions against respective ground truths

The decoder Dec_ϕ (prompt encoder + mask decoder) is shared between both passes and adapted via LoRA [Hu et al., 2022] with rank 8, yielding 1.9M trainable parameters (1.89% of total 100.5M). The image encoder remains frozen, so the second forward pass reuses the cached embedding $\mathbf{e}$ with minimal overhead.

Loss Function. The total loss combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_s \mathcal{L}_{\text{switch}} + \lambda_g \mathcal{L}_{\text{ground}} \quad (3)$$

The segmentation loss averages Dice+BCE on both predictions:

$$\mathcal{L}_{\text{seg}} = \frac{1}{2} [\ell(\hat{y}_A, y_A) + \ell(\hat{y}_B, y_B)] \quad (4)$$

The switch margin loss enforces output separation:

$$\begin{aligned} \mathcal{L}_{\text{switch}} = & [m + D(\hat{y}_A, y_B) - D(\hat{y}_A, y_A)]_+ \\ & + [m + D(\hat{y}_B, y_A) - D(\hat{y}_B, y_B)]_+ \end{aligned} \quad (5)$$

where $D(\cdot, \cdot)$ is soft Dice similarity, $m=0.2$ is the margin, and $[\cdot]_+ = \max(0, \cdot)$. This loss penalizes predictions that are more similar to the wrong organ than to the correct one, maintaining a persistent gradient signal throughout training.

The grounding loss ensures prompt points lie inside predictions:

$$\mathcal{L}_{\text{ground}} = -\log \sigma(\hat{y}_A[p_A]) - \log \sigma(\hat{y}_B[p_B]) \quad (6)$$

where σ is the sigmoid function and $\hat{y}[p]$ denotes the logit at the prompt location.

We use $\lambda_s=1.0$ and $\lambda_g=0.5$ throughout. As our ablation demonstrates (§4.4), the specific loss weights matter far less than the paired training paradigm itself.

Key Insight. The critical mechanism is not any particular loss function but the *paired prompt exposure*: by presenting two different prompts for the same image in a single training step, the model cannot satisfy the objective by ignoring prompts. It must learn to condition its output on the prompt to minimize loss on both predictions simultaneously.

4 Experiments

4.1 Experimental Setup

Datasets. We use AMOS [Ji et al., 2022] (240 training, 60 validation CT volumes, 13 abdominal organs, resampled to 1.5mm isotropic, 128^3 crops) as the primary dataset and BTCV [Landman et al., 2015] (21 training, 9 validation, 13 organs) for external validation. Both datasets provide multi-organ annotations enabling paired prompt sampling.

Model and Adaptation. SAM-Med3D-turbo [Wang et al., 2023] with frozen ViT-B image encoder. LoRA adaptation (rank=8) on prompt encoder and mask decoder, yielding 1.9M trainable parameters (1.89% of 100.5M total).

Training. 50 epochs, AdamW optimizer (LR 10^{-4} , weight decay 0.01), 2-epoch linear warmup followed by cosine decay to 10^{-6} , batch size 1 with gradient accumulation 4. Three random seeds (42, 123, 7) for main experiments. Checkpoint selection: maximum validation Dice subject to Switch Accuracy ≥ 0.90 .

Evaluation Protocol. We construct a fixed prompt bank of 514 paired prompts (deterministic across all methods) to ensure fair comparison. Each pair consists of two point prompts targeting different organs in the same volume. We report Dice score and Switch Accuracy with McNemar test for statistical significance ($p < 0.001$).

Baselines. (1) **Frozen:** SAM-Med3D without adaptation; (2) **seg-only:** LoRA fine-tuning with standard Dice+BCE loss, single prompt per step; (3) **Ours:** Paired Prompt-Contrastive Training with full loss (Eq. 3).

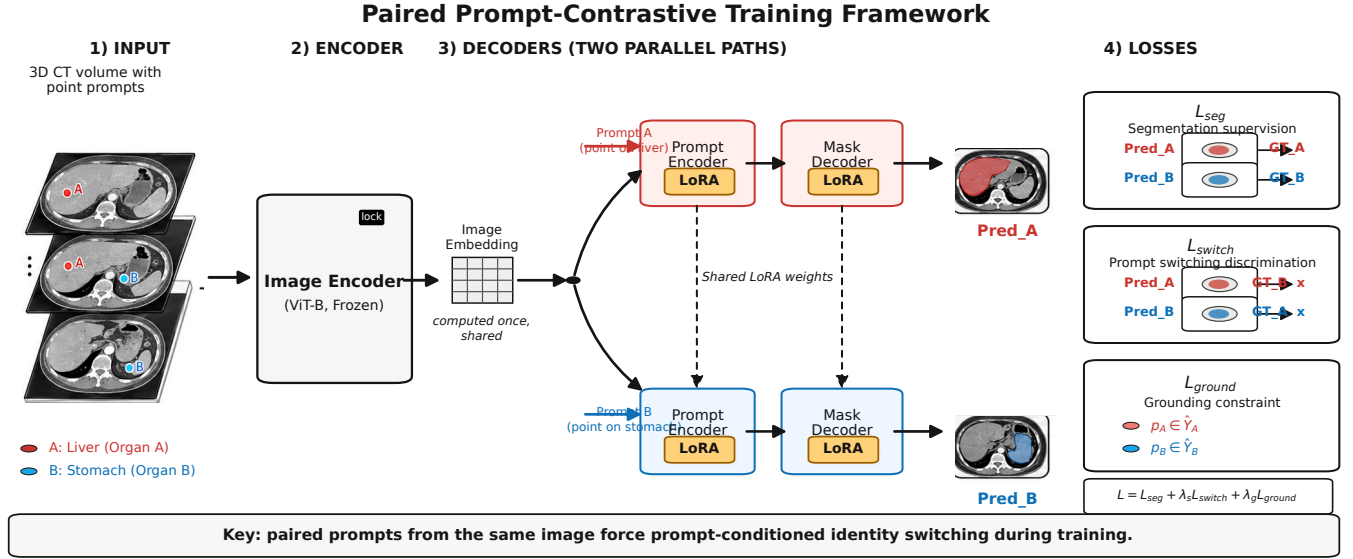


Figure 2: Paired Prompt-Contrastive Training framework. The frozen image encoder computes a shared embedding. Two different organ prompts are decoded through the same LoRA-adapted decoder, producing two predictions supervised by their respective ground truths. The switch margin loss enforces that each prediction is closer to its target organ than to the alternative.

Table 1: Main results on AMOS and BTCV. Our method achieves near-perfect Switch Accuracy with modest Dice cost. McNemar test: $p < 0.001$ for Switch improvement across all seeds.

Method	Dice	Switch Acc.
<i>AMOS (3 seeds, mean $\pm$ std)</i>		
Frozen SAM-Med3D	0.450	0.208
seg-only (LoRA)	0.720 ± 0.006	0.483 ± 0.037
Ours (Paired)	0.676 ± 0.006	0.993 ± 0.012
<i>BTCV (external validation)</i>		
seg-only (LoRA)	0.745	0.312
Ours (Paired)	0.704	1.000

4.2 Main Results

Table 1 presents the main comparison on AMOS (3 seeds) and BTCV (external validation).

Standard fine-tuning (seg-only) improves Dice substantially (0.450 $\rightarrow$ 0.720) but Switch Accuracy only reaches 0.483—the model still ignores prompts nearly half the time. Our method achieves Switch Accuracy 0.993 (+51.0 points over seg-only) with a 4.4-point Dice cost. The Dice gap is modest and expected: some segmentation capacity is redirected toward prompt-conditioned behavior rather than unconditional organ prediction.

On BTCV external validation, the improvement is even larger: Switch 1.000 vs 0.312 (+68.8 points), demonstrating strong generalization. The consistent pattern across datasets confirms that paired training fundamentally changes model behavior rather than overfitting to dataset-specific patterns.

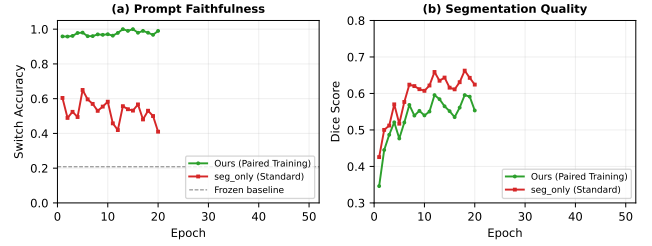


Figure 3: Training dynamics over 50 epochs. (a) Switch Accuracy: our method maintains >0.95 throughout while seg-only stagnates at ~ 0.5 . (b) Dice: both methods improve, with seg-only reaching slightly higher final Dice due to unconstrained optimization.

4.3 Training Dynamics

Figure 3 shows Dice and Switch Accuracy over 50 training epochs. The seg-only baseline exhibits a characteristic pattern: Dice climbs steadily while Switch Accuracy fluctuates around 0.5 without improvement—the model learns better segmentation but not better prompt following. In contrast, our method maintains Switch Accuracy above 0.95 from epoch 1 onward while Dice gradually improves, demonstrating that paired training establishes prompt faithfulness immediately and maintains it throughout optimization.

4.4 Mechanism Ablation

To isolate the contribution of each component, we conduct a factor-isolation experiment (Table 2, 20 epochs, seed=42). We vary two binary factors—paired vs. single prompt training, and presence vs. absence of auxiliary losses—to determine what drives faithfulness.

The results reveal a clear hierarchy of importance:

Table 2: Mechanism ablation (20 epochs, AMOS). Paired training is the dominant factor (+40pt Switch). Grounding without paired training is counterproductive.

Method	Paired	Sw. Acc	Dice
seg-only	No	0.55	0.62
+ grounding only	No	0.43	0.62
paired (seg only)	Yes	0.94	0.60
+ grounding	Yes	0.99	0.57
+ ground + switch	Yes	0.99	0.60

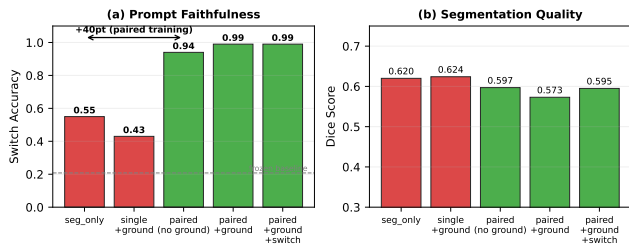


Figure 4: Mechanism ablation visualization. Single-prompt methods (red) fail regardless of loss design. Paired-prompt methods (green) succeed regardless of auxiliary losses. The gap between groups (~ 40 points) dwarfs within-group variation.

Paired training is necessary and sufficient (+40 points). Moving from single-prompt to paired-prompt training—even without any auxiliary loss—jumps Switch Accuracy from 0.55 to 0.94. This is the dominant mechanism.

Grounding adds marginal benefit on top of paired training (+5 points, 0.94 $\rightarrow$ 0.99). However, grounding *without* paired training actually *decreases* Switch Accuracy (0.55 $\rightarrow$ 0.43). This counterintuitive result suggests that grounding reinforces the model tendency to predict based on spatial priors without learning to differentiate between prompts.

Switch margin loss adds negligible benefit (0.99 $\rightarrow$ 0.99). When paired training already provides the contrastive signal, the explicit margin loss is redundant.

Figure 4 visualizes this hierarchy, making clear that the contribution is a *training paradigm* (paired exposure), not a specific loss function.

4.5 Hyperparameter Robustness

Table 3 demonstrates that all paired-training configurations maintain Switch Accuracy ≥ 0.978 across a wide range of margin values and loss weights.

The Dice score varies modestly with λ_{switch} (0.545–0.595), suggesting a mild tradeoff: higher switch weight slightly reduces segmentation accuracy. However, Switch Accuracy remains near-perfect across all settings, indicating that the paired training mechanism is inherently robust and does not require careful tuning.

Table 3: Hyperparameter robustness (20 epochs, AMOS). All paired-training configurations maintain Switch ≥ 0.978 .

Configuration	Switch Acc.	Dice
margin = 0.1	0.990	0.561
margin = 0.2 (default)	0.990	0.595
margin = 0.3	0.990	0.555
$\lambda_{\text{switch}} = 0.25$	0.989	0.567
$\lambda_{\text{switch}} = 1.0$ (default)	0.990	0.595
$\lambda_{\text{switch}} = 2.0$	0.990	0.545

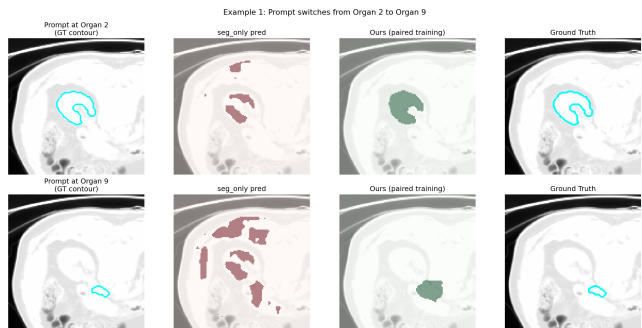


Figure 5: Qualitative comparison. Each row shows predictions for a different prompt target on the same image. seg-only produces identical outputs regardless of prompt (unfaithful). Our method correctly segments the prompted organ (faithful).

4.6 Cross-Domain Analysis

To assess whether prompt unfaithfulness is specific to 3D medical SAM, we evaluate SAM2 [Ravi et al., 2024] on COCO natural images using the same Switch Accuracy protocol. SAM2 achieves 0.954 Switch Accuracy out of the box, indicating that prompt unfaithfulness is specific to 3D medical adaptation—not a general foundation model problem.

We hypothesize this specificity arises from two factors: (1) limited diversity in medical training data (anatomical structures are highly consistent across patients), enabling the model to exploit spatial priors; and (2) the frozen encoder paradigm, where only the decoder is adapted, limiting the model capacity to learn new prompt-conditioned features.

4.7 Qualitative Results

Figure 5 shows representative examples comparing seg-only and our method. When given two different organ prompts on the same image, seg-only produces nearly identical predictions (both targeting the dominant organ), while our method correctly switches the segmentation target to match each prompt.

5 Analysis and Discussion

5.1 Why Paired Training Works

The effectiveness of paired prompt training can be understood through the lens of counterfactual supervision. In single-prompt training, the model observes (x, p_A, y_A) and learns

Table 4: Computational overhead analysis. Paired training adds minimal cost during training and zero cost at inference.

Metric	seg-only	Ours
Total parameters	100.5M	100.5M
Trainable parameters	1.9M (1.89%)	1.9M (1.89%)
Training time/epoch	1117s	1177s (+5.3%)
Total training (50ep)	6.2h	6.5h
Inference time	identical	identical
Checkpoint size	7.6MB	7.6MB

$f(x, p_A) \approx y_A$. However, nothing prevents the model from learning $f(x, \cdot) \approx y_A$ for *all* prompts—the training signal is consistent with prompt-independent behavior. Paired training adds the simultaneous constraint $f(x, p_B) \approx y_B$, which is only satisfiable if the model actually uses the prompt to differentiate its output.

This is analogous to the distinction between correlation and causation in causal inference: single-prompt training establishes correlation between prompts and outputs (they co-occur), while paired training establishes a causal relationship (changing the prompt *causes* the output to change).

5.2 The Dice-Switch Tradeoff

Our method incurs a 4.4-point Dice cost compared to seg-only. This tradeoff is inherent: a model that always predicts the largest organ achieves high average Dice (by being correct for the dominant class) but zero Switch Accuracy. Conversely, a perfectly faithful model must sometimes predict smaller, harder organs when prompted, reducing average Dice. The 4.4-point gap represents the cost of controllability—a modest price for transforming an uncontrollable model into a reliable interactive tool.

We note that the absolute Dice of 0.676 is achieved on a challenging fixed prompt bank that includes all organ pairs (including small organs like adrenal glands). Per-organ analysis shows that large organs (liver, spleen) maintain Dice >0.85 while the gap concentrates on small, anatomically challenging structures.

5.3 Computational Efficiency

Table 4 summarizes the computational overhead. The second decoder forward pass adds only 5.3% training time because the expensive image encoder computation is shared. At inference, only a single prompt is processed (standard interactive use), so there is zero overhead.

5.4 Limitations

Single architecture. We validate on SAM-Med3D only. While the mechanism (paired exposure) is architecture-agnostic, empirical validation on SegVol or other 3D medical SAMs would strengthen generality claims.

Dice cost. The 4.4-point Dice gap, while modest, exists. Applications requiring maximum segmentation accuracy without interactive control may prefer the seg-only baseline.

Annotation requirement. Paired prompt sampling requires multi-organ annotations in the same volume. This is available in standard benchmarks (AMOS, BTCV) but may limit applicability to single-organ datasets.

Point prompts only. We evaluate point-prompt faithfulness. Extension to box prompts and text prompts (as in SegVol) remains future work.

6 Conclusion

We identify prompt unfaithfulness as a hidden failure mode in 3D medical Segment Anything Models: these models achieve acceptable segmentation Dice while largely ignoring user prompts (Switch Accuracy 20.8%). This failure is distinct from prompt robustness, hidden by standard evaluation metrics, and catastrophic for interactive clinical use.

We propose Switch Accuracy as a diagnostic metric that exposes this failure, and trace the root cause to single-prompt training, which never constrains counterfactual prompt behavior. Our solution—Paired Prompt-Contrastive Training—exposes the model to two different prompts per image during training, forcing prompt-conditioned output differentiation.

Comprehensive mechanism ablation reveals that paired prompt exposure is the dominant factor (+40 points Switch Accuracy), while specific loss designs provide only marginal additional benefit. The method achieves 99.3% Switch Accuracy (from 20.8% baseline) with 4.4-point Dice cost, generalizes across datasets (BTCV: Switch 1.000), and adds negligible overhead (5.3% training time, zero inference cost).

Our findings suggest that prompt faithfulness should be a standard evaluation criterion for interactive segmentation models, alongside traditional accuracy metrics. The paired training paradigm is simple, effective, and broadly applicable to any promptable model where controllability is essential.

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